

AR Sandbox: Dynamic Terrain Mapping and Water Simulation in an AR Environment

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Executive Summary

The Augmented Reality (AR) Sandbox project integrates 3D visualization with a physical medium to create a dynamic, interactive tool for topographic and hydrologic education. By combining an Xbox 360 Kinect sensor, a BenQ projector, and open-source real-time fluid dynamics software from UC Davis, the system allows users to manipulate physical sand to create terrain that is instantly augmented with a digital elevation map and simulated water flow. This project bridges the gap between abstract cartographic theory and tangible learning, providing a powerful platform for demonstrating how geographic features interact with rainfall in an intuitive, hands-on environment. While the standard UC Davis implementation typically utilizes a smaller footprint, this project successfully scaled the simulation to a larger 48" x 36" sandbox to enhance collaborative use and visual impact.

Background

The AR Sandbox bridges the gap between complex environmental data and public understanding of water science. Traditional 2D maps often fail to convey the dynamic nature of watersheds and drainage patterns to non-experts. By providing an interactive interface, this project allows users to visualize the immediate impact of land manipulation on water flow in real time. This approach increases accessibility to water science, moving it from the lab into a shared space where anyone can see how geography dictates rainfall movement. Addressing this need is crucial for helping communities understand flood risks, resource conservation, and the effects of urban development. Ultimately, the project simplifies the invisible forces of hydrology into an intuitive, hands-on experience.

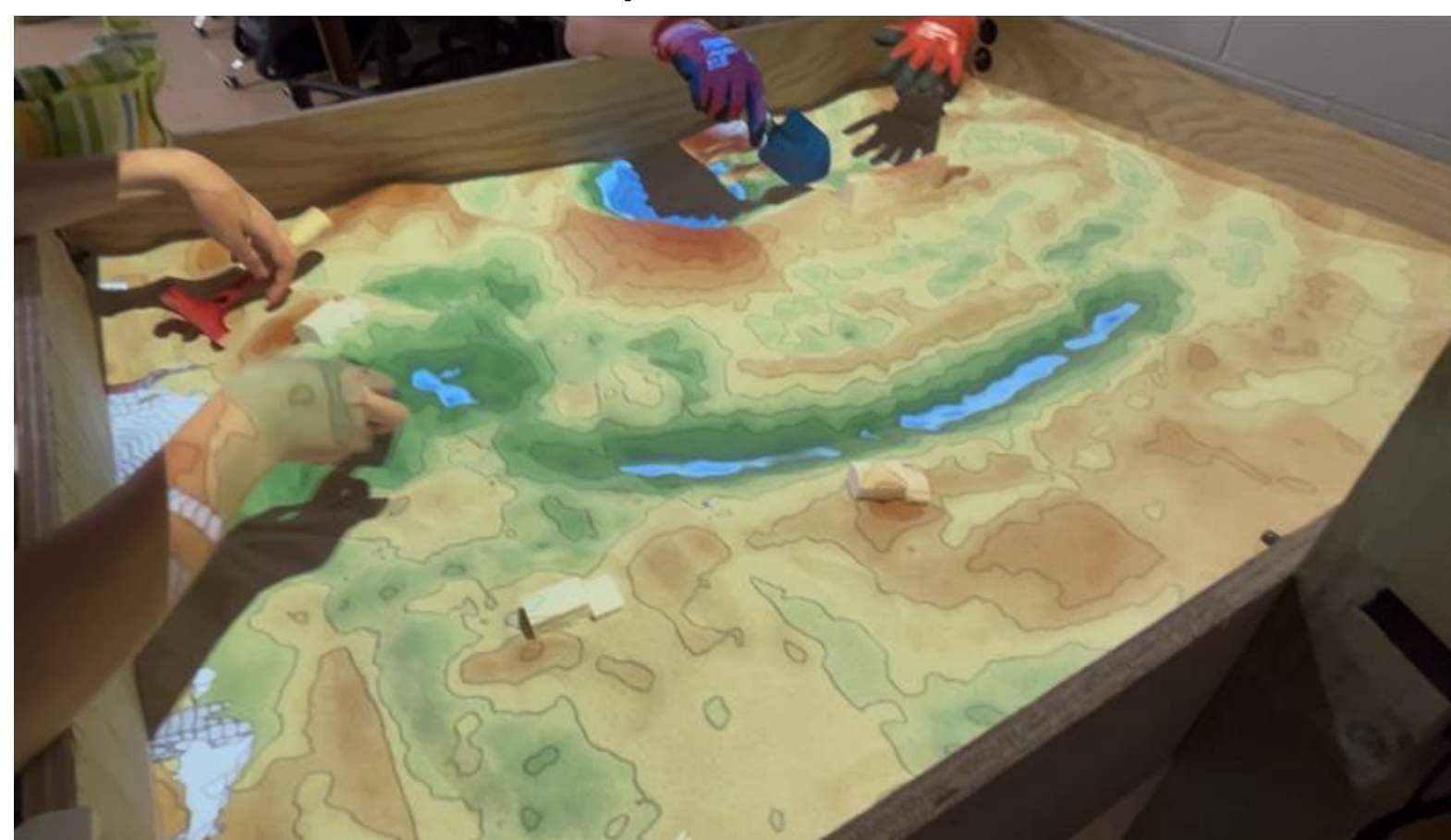


Figure 1. Children playing in the AR sandbox

Experimental Setup

The project's hardware configuration is designed to bridge physical manipulation with high-speed digital rendering. The core setup features a Xbox 360 Kinect 3D camera mounted above the sandbox to scan surface depth, paired with a BenQ short-throw projector to overlay the processed topographic map and fluid simulations directly onto the sand. Driven by a workstation operating on Linux, the system utilizes VRUI VR development toolkit and open-source software from UC Davis to convert raw depth data into a real-time digital elevation model. A key mechanical enhancement in this design was scaling the physical sandbox from the standard 40" x 30" footprint to a larger 48" x 36" structure, providing more surface area for complex watershed modeling.

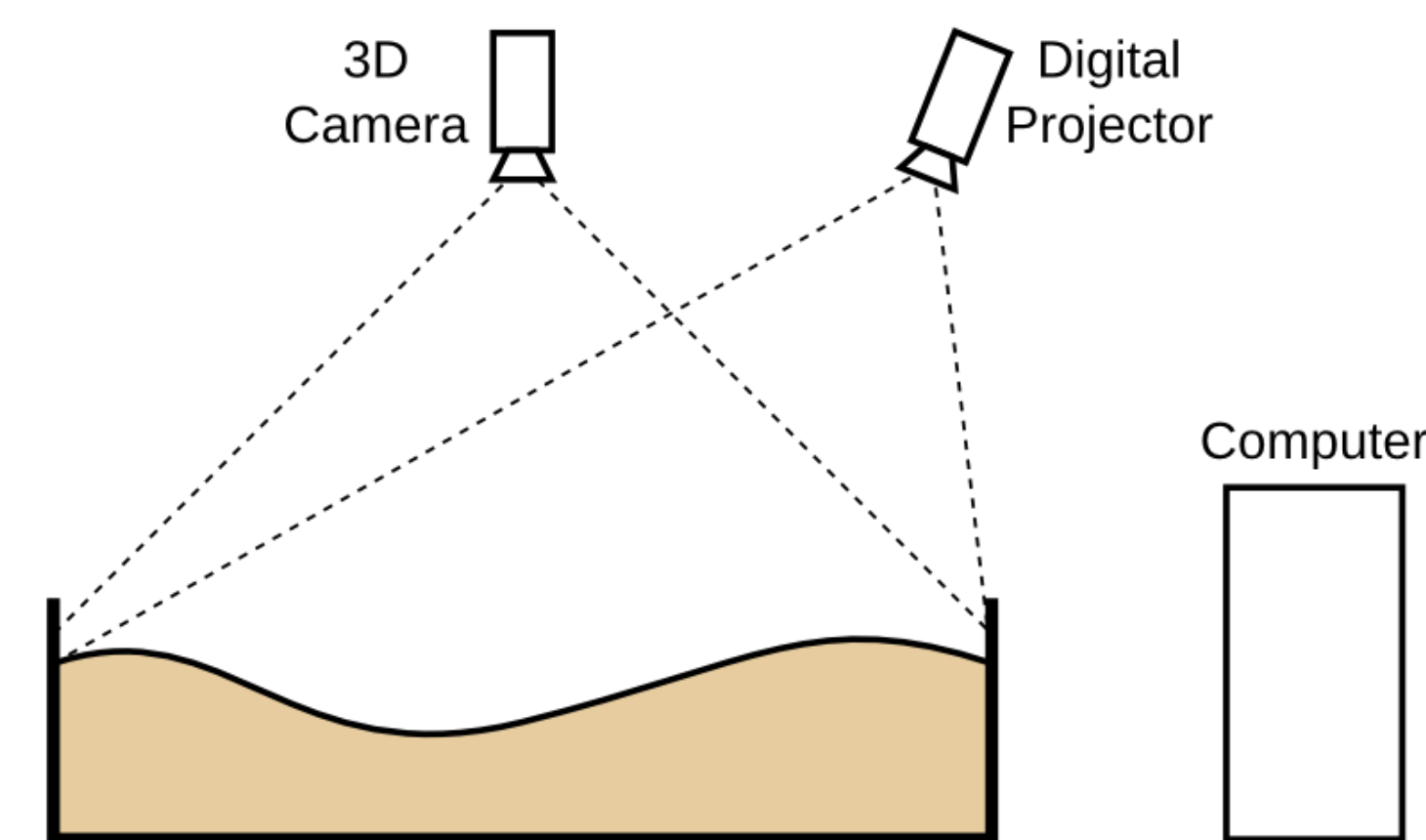


Figure 2. Diagram showing AR Sandbox setup

Experimental Test Plan

The software infrastructure utilizes the VRUI VR Toolkit and the Kinect 3D Video Package to translate raw infrared depth streams into a responsive digital elevation model. The process begins with a tie-point calibration, where projected cross-hairs are matched to physical coordinates at varying heights. This allows the software to calculate a transformation matrix that aligns the Kinect's 3D camera space with the projector's 2D coordinate system, ensuring digital contour lines accurately overlay the physical sand.

Once calibrated, the software assigns color-coded elevation bands and simulates real-time fluid dynamics using depth-integrated Navier-Stokes equations. We validate the system by triggering "virtual rain" with hand gestures, ensuring the software correctly interprets the sensor data to generate realistic water flow and pooling. To accommodate the expanded 48" x 36" footprint, we adjusted the sea level and per-pixel scale settings, maintaining realistic vertical exaggeration across the larger terrain and effectively bridging the gap between physical materials and digital simulation.

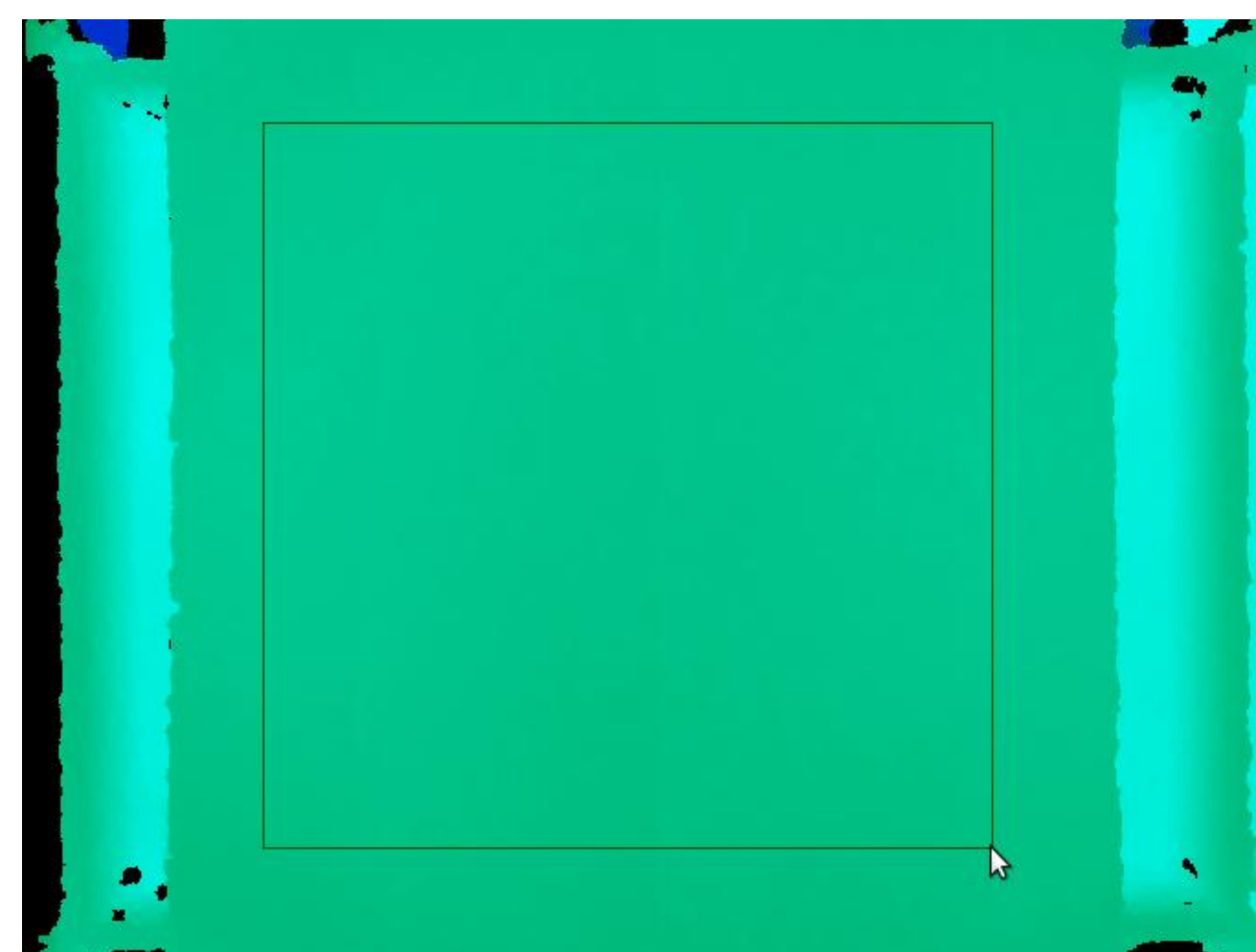


Figure 3. Raw Kinect Viewer point of view calculating plane equation for AR Sandbox

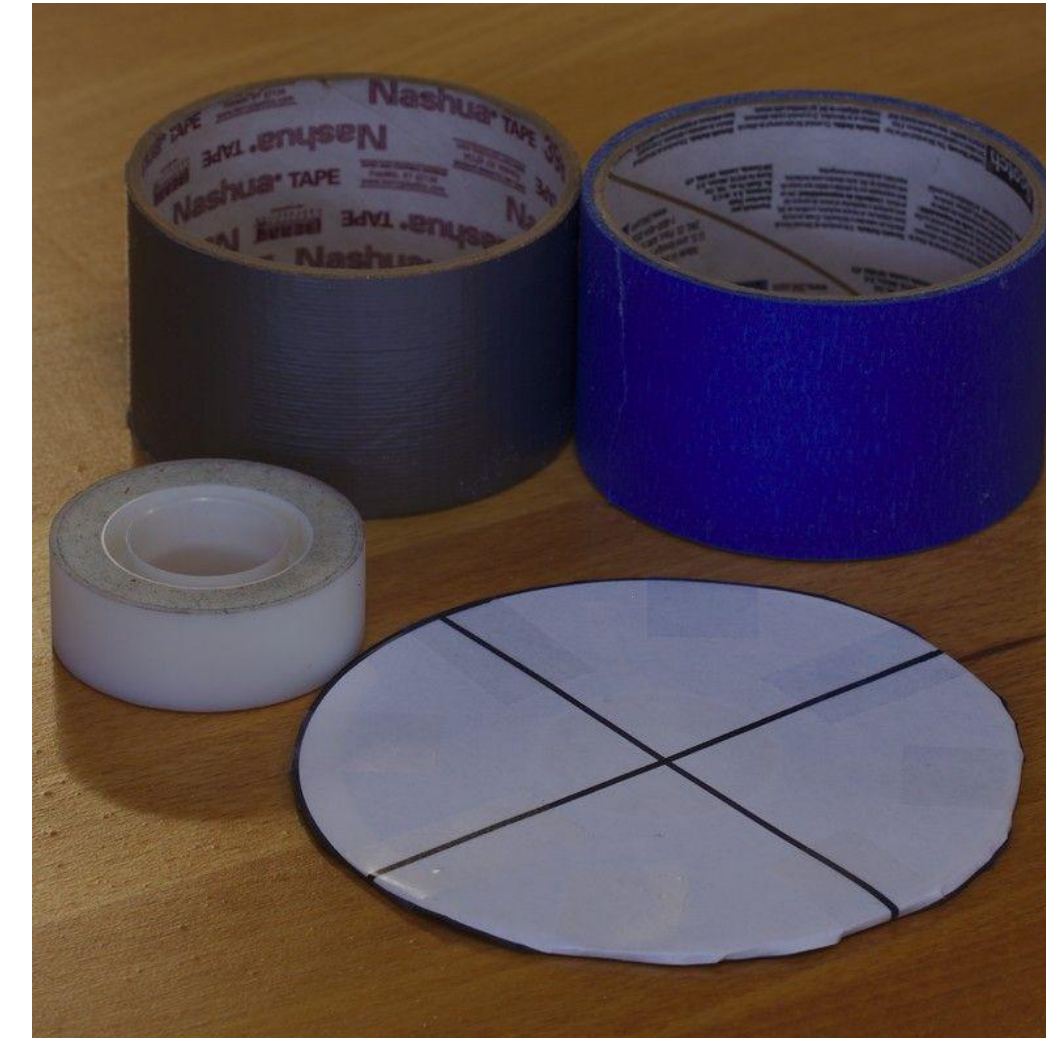


Figure 4. Items used for tie point measurement method

Experimental Results

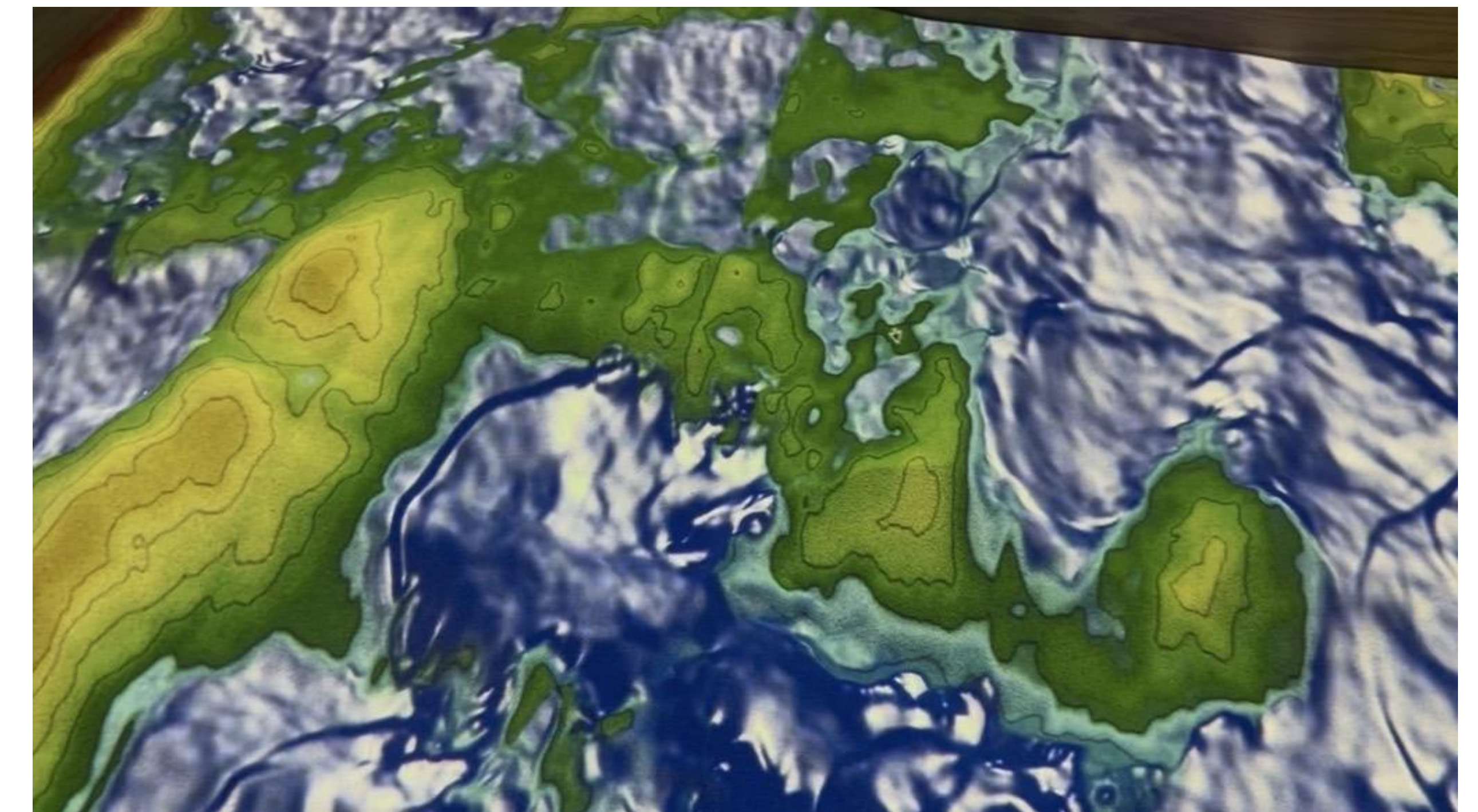


Figure 5. AR Sandbox with water simulation in use

Conclusions

Our design successfully fulfills the project's goal of creating an immersive, interactive topography tool that bridges the gap between physical materials and digital simulations. By scaling the sandbox to a 48" x 36" footprint, we enhanced the system's capacity for collaborative learning, allowing multiple users to engage with the terrain simultaneously. Key insights gained during development include the critical importance of high-visibility sand for projection clarity and the necessity of precise "tie-point" calibration to ensure that digital contour lines and physical peaks align seamlessly. The project demonstrates that tangible computing can significantly improve the accessibility of water science and topographic education by making abstract concepts visible and touchable. Looking ahead, we aim to expand the system's analytical capabilities by developing a function to measure simulated rainfall in both imperial and metric units, providing users with quantitative data to match the visual simulation. We would like to acknowledge the UC Davis W.M. Keck Center for Active Visualization in the Earth Sciences for providing the open-source software and documentation that served as the foundation for this work.

References

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2. Reed, S., et al., "Shaping Watersheds Exhibit: An Interactive, Augmented Reality Sandbox for Advancing Earth Science Education," American Geophysical Union (AGU) Fall Meeting 2014, 2014, Abstract no. ED34A-01.
3. Reed, S., et al., "Augmented Reality Turns a Sandbox into a Geoscience Lesson," EOS 97, 2016, [Online]. Available: <https://doi.org/10.1029/2016EO056135>

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